LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	NACA series, Vortex theory, Magnus effect, Kutta Joukowski law, Span wise flow, down Wash wing tip and trailing edge vortices
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	NACA series, Vortex theory, Magnus effect, Kutta Joukowski law, Span wise flow, down Wash wing tip and trailing edge vortices. Methods for preventing span-wise flow.
5. Specific Objectives	11.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of NACA profile charts and Vortex formation diagrams using PPT slides. Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aerofoil section?
(b) Define the term 'span-wise flow' in the context of a wing.
(c) Recall the relationship between circulation and lift force.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• NACA series classification and profile characteristics • Conditions of performance for standard aerofoils • Vortex theory fundamentals and circulation concepts • Magnus effect application in rotating cylinders • Kutta Joukowski law derivation and lift calculation	L & Ds	PPT, IB	15 min
(b)	• Formation of span-wise flow and downwash patterns • Wing tip vortices generation and trailing edge effects • Induced drag consequences of vortex systems • Methods for preventing span-wise flow • Design features to mitigate downwash and vortices	L & Ds	PPT, PDF	20 min
(c)	• Summary of lift theories and vortex interactions • Practical implications for helicopter rotor performance • Review of NACA series selection criteria • Connection between Kutta condition and flow separation • Overview of span-wise flow control techniques	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the Kutta Joukowski law relates circulation to lift force.
(b) Describe the mechanism by which wing tip vortices induce drag.
(c) List two methods used to prevent span-wise flow on a wing.

12. Home Assignment and Follow Up	Prepare a brief note comparing the NACA 4-digit and 5-digit series profiles, highlighting their differences in camber and thickness distribution for the next lesson on drag analysis.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Drag analysis, induced, parasite, form, skin friction, profile and total drag. D-V curve.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min